Hints of the solution-Final - 2026

1. (a) Here $F(x)$ is continuous and increasing and $\lim_{x\to\infty} F(x) = 1$, $\lim_{x\to-\infty} F(x) = 0$. It is continuous. Unless you show all the properties you will not get credit.

$$f(x) = \lambda \int_0^\infty \frac{1}{u}\phi\left(\frac{x-u}{u}\right) e^{-\lambda u} du.$$

   (b) Note that if U is $\exp(\lambda)$ and $X|U = u$ is $N(u, u^2)$, then X has the CDF $F(x)$. Hence, $X = U + UZ$, where $U \sim$ Exp(λ) and $Z \sim$ N(0,1) and they are independent.

2. Note that if $X \sim$ Gamma(α, λ, μ), then $X = Y + \mu$, where $Y \sim$ Gamma(α, λ). Now $E(Y^m) = \dfrac{\Gamma(m+\alpha)}{\Gamma(\alpha)\lambda^m}$. Therefore, $A = \alpha+\mu$ and $B = \alpha(\alpha+1)+2\alpha\mu+\mu^2$. $\widehat{\alpha} = (B - A^2)$, $\widehat{\mu} = A - B + A^2$

3. The log-likelihood is

$$l(\alpha, \lambda) = n\alpha \ln \lambda - n \ln \Gamma(\alpha) + (\alpha - 1)A - B\lambda.$$

   Hence, $\widehat{\lambda}(\alpha) = \dfrac{n\alpha}{B}$. Therefore,

$$g(\alpha) = n\alpha(\ln n + \ln \alpha - 1 - \ln B) - n \ln \Gamma(\alpha) + (\alpha - 1)A.$$

   and $\dfrac{\widehat{\alpha}}{\widehat{\lambda}} = B/n$.

4. We have done it in the class. If $U_1 \sim$ Exp(λ_1), $U_2 \sim$ Exp(λ_2), then $E(U_1|U_1 > c) = c + \dfrac{1}{\lambda_1}$ and $E(U_1|\min\{U_1, U_2\} = c) = c \times \dfrac{\lambda_1}{\lambda_1+\lambda_2} + \left(c + \dfrac{1}{\lambda_1}\right) \times \dfrac{\lambda_2}{\lambda_1+\lambda_2} = c + \dfrac{1}{\lambda_1} \times \dfrac{\lambda_2}{\lambda_1+\lambda_2}$.

5. The log-likelihood function without the additive constant becomes

$$l(\alpha, \beta) = -7\alpha - 28\beta + \sum_{i=1}^{7} x_i \ln(\alpha + i\beta) = -7\alpha - 28\beta + c\sum_{i=1}^{7} \ln(\alpha + i\beta)$$

   If $\beta = 1$, $g(\alpha) = c\sum_{i=1}^{7}(\alpha + i)^{-1} - 7$. If $\alpha = 0$, $\widehat{\beta} = \dfrac{c}{4}$.

6. Here $\widehat{p} = \#$ of negative values/7. $\widehat{\lambda} = \dfrac{7}{\sum_{i:x_i<0} x_i^2 + 2\sum_{i:x_i>0} x_i^2}$.

7. Here $F(x) = x - \theta + 1/2$ if $\theta - 1/2 < x < \theta + 1/2$, 0, if $x < \theta - 1/2$ and 1, if $x > \theta + 1/2$. Hence one needs to maximize

$$g(\theta) = \ln\left(1 - x_{(n)} + \theta - \frac{1}{2}\right) + \ln\left(x_{(1)} - \theta + \frac{1}{2}\right)$$

   with respect to θ to compute the MPS estimator of θ. $\widehat{\theta} = (x_{(1)} + x_{(n)})/2$.

8.

$$
\begin{aligned}
f_1(x,y) &= \alpha^2\lambda_1(\lambda_2+\lambda_3)x^{\alpha-1}y^{\alpha-1}e^{-\lambda_1 x^\alpha-(\lambda_2+\lambda_3)y^\alpha}\\
f_2(x,y) &= \alpha^2(\lambda_1+\lambda_3)\lambda_2 x^{\alpha-1}y^{\alpha-1}e^{-(\lambda_1+\lambda_3)x^\alpha-\lambda_2 y^\alpha}
\end{aligned}
$$

9. The log-likelihood function of α and δ is

$$
l(\alpha,\delta) = 10\ln\alpha + \sum_{i=1}^{10}\ln\left(1+\frac{t_i}{\delta}\right) - \alpha\left(\sum_{i=1}^{10}t_i + \frac{1}{\delta}\sum_{i=1}^{10}t_i^2\right).
$$

Hence, $\widehat{\alpha}(1) = \dfrac{10\delta}{\delta\sum_{i=1}^{10}t_i + \sum_{i=1}^{10}t_i^2}$.

$$
\begin{aligned}
l(1,1) &= \sum_{i=1}^{10}\ln(1+t_i) - \left(\sum_{i=1}^{10}t_i + \sum_{i=1}^{10}t_i^2\right) = -13(-49)\\
l(1,2) &= \sum_{i=1}^{10}\ln(1+0.5t_i) - \left(\sum_{i=1}^{10}t_i + \frac{1}{2}\sum_{i=1}^{10}t_i^2\right) = -11(-33)\\
l(2,1) &< 10 + \sum_{i=1}^{10}\ln(1+t_i) - 2\left(\sum_{i=1}^{10}t_i + \sum_{i=1}^{10}t_i^2\right) = -23(-99)\\
l(2,2) &< 10 + \sum_{i=1}^{10}\ln(1+0.5t_i) - 2\left(\sum_{i=1}^{10}t_i + \frac{1}{2}\sum_{i=1}^{10}t_i^2\right) = -16(-63)
\end{aligned}
$$